

COSC 594

DM for Robotic Planning (I)

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Outline

- MOTIVATION
- ROBOTIC PLANNING
- ENERGY-BASED PLANNING

Outline

■ MOTIVATION

Motivation: why robot learning?

■ CLASSICAL ROBOTICS



Automation of foundry with robot

- ★ precise model
- ★ specific task
- ★ static environment
- ★ expensive and slow design

Motivation: why robot learning?

■ END-TO-END ROBOTICS



Gemini Robotics 1.5

- ★ learn from data/experience
- ★ multiple tasks
- ★ dynamic environment
- ★ cost-effective and quick design

Outline

■ MOTIVATION

■ ROBOTIC PLANNING

Trajectory planning

System dynamics: $s_{t+1} = f(s_t, a_t)$



Trajectory $\tau = (s_0, a_0, s_1, a_1, \dots, s_T, a_T)$

■ REQUIREMENTS

- ★ energy efficiency
- ★ obstacle avoidance
- ★ trajectory smoothness

Trajectory optimization

■ ACTION OPTIMIZATION

$$\underset{\tau}{\text{maximize}} \quad R(\tau; \textcolor{blue}{s}_0) \quad := \quad \sum_{t=0}^T r(s_t, a_t)$$

reward, e.g., negative cost of collision

- ★ an optimal action sequence $a_{0:T}^*$
- ★ generate an optimal trajectory τ^*

$$\tau^* \quad = \quad (\textcolor{blue}{s}_0, a_0^*, s_1^*, a_1^*, \dots, s_T^*, a_T^*)$$

- ★ task info $\textcolor{blue}{h}$, e.g., s_0, s_T

Planning as inference

Random trajectory: $\tau = (s_0, a_0, s_1, a_1, \dots, s_T, a_T)$

Posterior trajectory distribution: $p(\tau | h) \propto p(h | \tau)p(\tau)$

$p(h | \tau)$ – likelihood of completing tasks

$p(\tau)$ – prior

■ MAXIMUM A POSTERIORI

maximize _{τ} $p(\tau | h)$



maximize _{τ} $\log p(\tau | h)$

Recovering trajectory optimization

Random trajectory: $\tau = (s_0, a_0, s_1, a_1, \dots, s_T, a_T)$

Energy-based distribution: $p(h | \tau) \propto e^{E(\tau; h)}$ w/ $E(\tau; h) = R(\tau; h)$

Prior distribution: $p(\tau) = \text{Unif}$

■ POSTERIORI AS LIKELIHOOD

$$\underset{\tau}{\text{maximize}} \quad \log p(\tau | h)$$



$$\underset{\tau}{\text{maximize}} \quad \log p(h | \tau)$$



$$\underset{\tau}{\text{maximize}} \quad R(\tau; h)$$

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Energy-based modeling of trajectory distribution

Energy-based model: $p_{\theta}(h \mid \tau) \propto e^{E_{\theta}(\tau; h)}$ w/ energy $E_{\theta}(\tau; h)$

Trajectory data: $\mathcal{D} = \{(\tau^k, h^k)\}_{k=1}^K$ w/ K modes

Prior distribution: $p(\tau) = \text{Unif}$

■ MULTI-MODAL LIKELIHOOD

$$p_{\theta}(h \mid \tau) \propto e^{E_{\theta}(\tau; h)} = \prod_{k=1}^K e^{E_{\theta^k}(\tau; h^k)}$$

- ★ multiple tasks
- ★ multiple modalities
- ★ different prior information

Incorporating prior

Energy-based model: $p_{\theta}(h \mid \tau) \propto \prod_{k=1}^K e^{E_{\theta^k}(\tau; h^k)}$

Prior distribution: $p(\tau)$

■ PRIOR DISTRIBUTION

$$p(\tau) \propto p(s_0) p(s_T) e^{\|\tau - \mu_{\tau}\|^2}$$

- ★ initial state $s_0 \sim p(s_0)$
- ★ final state $s_T \sim p(s_T)$
- ★ reference trajectory μ_{τ}

Sampling from energy-based model

Energy-based model: $p_{\theta}(h | \tau) \propto \prod_{k=1}^K e^{E_{\theta^k}(\tau; h^k)}$

Prior distribution: $p(\tau) \propto p(s_0) p(s_T) e^{\|\tau - \mu_{\tau}\|^2}$

■ POSTERIOR TRAJECTORY DISTRIBUTION

$$p(\tau | h) \propto p_{\theta}(h | \tau) \underbrace{p(\tau)}_{\text{Gaussian prior}} \propto e^{E_{\theta}(\tau; h)}$$

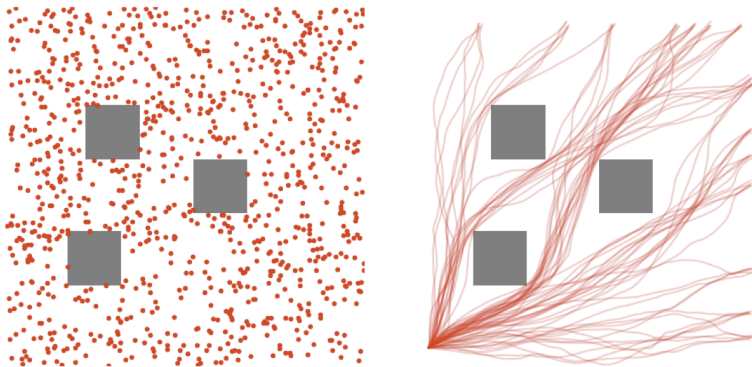
★ Langevin dynamics

$$\tau_t = \tau_{t-1} + \eta \nabla_{\tau} E_{\theta}(\tau_{t-1}; h) + \sqrt{2\eta} \epsilon_t \quad \text{where } \epsilon_t \sim \mathcal{N}(0, I)$$

As $t \rightarrow \infty$, dynamics converge to samples $\tau_t \sim p(\cdot | h)$

Experiment I: planar navigation

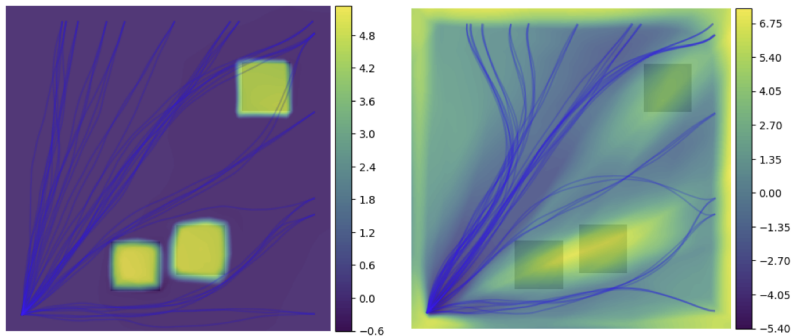
■ TWO SOURCES OF EMPIRICAL DATA



Random goal/target locations and three obstacles

Experiment I: planar navigation

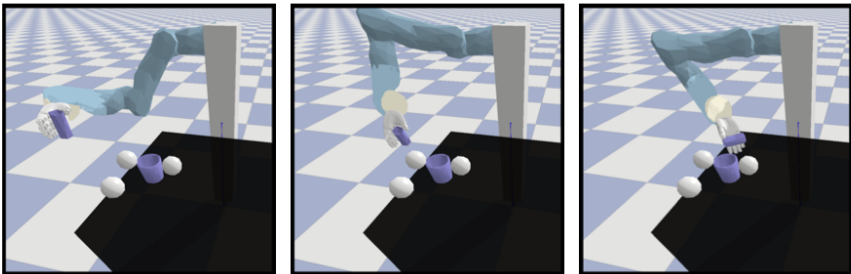
■ LEARNED TRAJECTORIES



Feasible trajectories w/ **new obstacles**

Experiment II: robot pouring

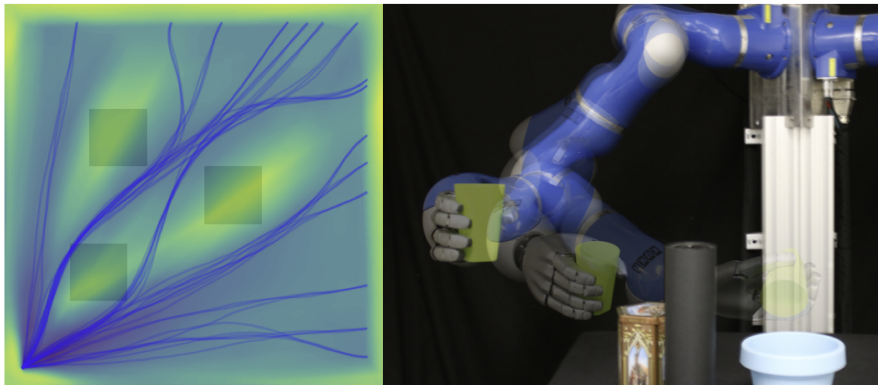
■ 3-D DATA



Trajectories from handcrafted policies

Experiment II: robot pouring

■ LEARNED TRAJECTORIES



Feasible trajectories in **new scene**

Summary

■ MOTIVATION

End-to-end robotics

■ ROBOTIC PLANNING

Trajectory optimization, planning as inference

■ ENERGY-BASED PLANNING

Energy-based modeling, incorporating prior